

KASILANKA BHOOPESH SIVA SRIKAR

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EDUCATION

Vellore Institute of Technology, Vellore, Tamil Nadu
Bachelor of Technology in Computer Science and Engineering

July 2022 – August 2026
CGPA: 8.53/10

WORK EXPERIENCE

Cloud, DevOps & Software Development Engineer Intern
Volteo Maritime India Pvt. Ltd. [🔗](#)

April 2025 – June 2025
Kakinada, Andhra Pradesh (Hybrid)

- Architected and managed **Kubernetes** clusters for **Wayship**, a maritime SaaS platform used by **2000+** seafarers across **200+** vessels; maintained **99.9% uptime** for real-time IoT telemetry processing handling **10M+** data points daily.
- Automated **CI/CD pipelines** using **Jenkins** and **Terraform** for containerized maritime systems, cutting deployment time by **40%** and eliminating manual configuration errors entirely.

PROJECTS

Healthcare AI with RAG and XAI | *FastAPI, ChromaDB, Hugging Face Transformers*

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- Developed a medical question-answering chatbot over 505K+ documents using hybrid RAG with dense retrieval, BM25, and Reciprocal Rank Fusion (RRF).
- Added explainability scores based on retrieval confidence, generation confidence, and source attribution.
- Wrote a 16-file pytest suite to test retrieval, generation, and explainability components, and set up a 6-stage CI/CD pipeline for linting, testing, security scanning, and deployment.

Study Flow: AI-Powered Productivity Suite | *React, FastAPI, PostgreSQL, Groq*

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- Created a full-stack study platform with a Pomodoro timer, Kanban board, code snippet manager, and activity heatmap, backed by 6 FastAPI routers.
- Integrated the Groq LLM API for 5 AI features, including study plan generation, code explanation, and focus-pattern analysis.
- Implemented Google OAuth and JWT-based authentication with PostgreSQL on Neon and automated Vercel deployments through GitHub CI/CD.

Handwritten Equation Solver | *TensorFlow, Next.js, FastAPI, OpenCV*

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- Engineered a CNN-based handwritten symbol classifier that recognizes 14 mathematical symbols with 95% test accuracy.
- Built an OpenCV preprocessing pipeline that converts input images to 32×32 grayscale images and segments individual symbols for inference.
- Developed a Next.js frontend and FastAPI inference API, deploying the frontend on Vercel and the backend on Hugging Face Spaces.

Headphone Safety | *Swift, CoreAudio, Audio Units*

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- Designed a native macOS menu bar application with 5 configurable headroom presets for limiting headphone output volume.
- Implemented a CoreAudio Audio Unit chain for real-time capture, limiting, and output, along with a virtual loopback driver for audio routing.
- Designed watchdog and rollback logic restoring safe output within 1 second of a device disconnect, with a startup recovery check guarding against unclean crashes.

TECHNICAL SKILLS

Languages: Python, Java, SQL, Bash, Swift

Frameworks & Libraries: FastAPI, React, Next.js, TensorFlow, OpenCV, LangChain

AI & LLM: RAG, ChromaDB, Hugging Face Transformers, Groq, MCP

Developer Tools: Linux, Git, Docker, GitHub Actions, Jenkins, Vim, VS Code

Databases: PostgreSQL, MySQL, Microsoft SQL Server

Cloud & Platforms: AWS, GCP, Vercel, Hugging Face Spaces, Qdrant Cloud, Neon